

Prashant Choudhary

Senior Research Fellow | PhD Research Scholar | Quantum Machine Learning & Quantum Computing

About

Senior Research Fellow(CSIR)/ PhD scholar at IIT (BHU), Varanasi, working on implementing Quantum Machine Learning (QML) for various real-life applications, developing new methods or algorithms for Quantum Architecture Search (QAS), finding the best decoders for Quantum Error Correction, and practical quantum software pipelines using Qiskit/PennyLane and Tensorflow/PyTorch Geometric.

Personal

Prashant Kumar Choudhary
Nationality: Indian
Location: Varanasi, India

Specialization

Quantum Machine Learning (QML)
Quantum Architecture Search (QAS)
Bayesian Optimization (BO)
Variational Quantum Circuits (VQC/QAOA-like)
Graph Neural Networks (GNNs)
Quantum Error Correction (QEC)

Tools & Stack

Python • PyTorch • PyTorch Geometric
Qiskit • PennyLane • NumPy
Git/GitHub • Colab • SLURM/HPC
LaTeX/Overleaf • Matplotlib

Interests

Quantum computing research, open-source tooling, scientific writing, and mentoring QML/QEC projects.

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RESEARCH SUMMARY

- **HQNN-FSP (Hybrid Quantum Neural Network for Financial Stock Prediction):** Developed a hybrid quantum-classical learning pipeline for financial time-series prediction, combining classical preprocessing/feature engineering with variational quantum circuits (VQCs) and classical neural components; evaluated predictive performance using standard forecasting metrics and performed baseline comparisons.
- **BO-driven Quantum Architecture Search (QAS) for VQCs:** Designed a sample-efficient search framework using Bayesian Optimization to discover high-performing variational circuits, including QAOA-like and template-based ansätze.
- **Graph-based surrogate modeling for circuits:** Represented quantum circuits as graphs and trained **GNN/MLP surrogates** to predict circuit performance and cost signals, enabling uncertainty-aware acquisition and faster exploration.
- **End-to-end reproducible research pipelines:** Built complete experimental workflows (data processing, circuit construction/mutation, black-box evaluation under noise, uncertainty calibration, baseline comparisons).
- **Current direction:** Exploring Quantum Error Correction (QEC) with an emphasis on practical constraints such as hardware-aware decoding latency and real-time classical processing.

EXPERIENCE & RESEARCH PROJECTS

2022–Present

PhD Research Scholar (Quantum Machine Learning)

IIT (BHU), VARANASI • India

Quantum Machine Learning and Quantum Computing research with emphasis on different methods for Quantum Architecture Search, Quantum Error Correction and surrogate modelings. Built reproducible pipelines using Qiskit/PennyLane and PyTorch Geometric; generated publication-ready plots/tables and artifacts (CSV/PDF/QASM), and performed systematic baseline comparisons.



2025–Present

Quantum Error Correction (QEC): Decoding & Hardware-Aware QEC Loop

ONGOING RESEARCH • QEC

Working on practical QEC workflows with focus on decoder design and evaluation under realistic noise, latency-aware classical processing constraints, and benchmarking of decoding approaches for near-term quantum architectures; building reproducible experiments and analysis pipelines for publishable results.



2024–2025

BO+VQC+GNN: Quantum Architecture Search Pipeline

RESEARCH PAPER • QAS & BO

Designed a graph-based Bayesian optimization framework to search VQC/QAOA-like circuit structures. Trained GNN surrogates over circuit graphs for performance prediction, used uncertainty-aware acquisition, ran comparisons against random/greedy/classical ML baselines, and automated export of figures/tables and QASM artifacts.



HQNN-FSP: Hybrid Quantum Neural Network for Financial Stock Prediction

RESEARCH PAPER • Quantum ML

Developed a hybrid quantum-classical forecasting pipeline for financial time-series prediction. Implemented variational quantum circuits integrated with classical neural components, designed data preprocessing + feature pipeline, and evaluated against classical baselines using standard forecasting metrics.



EDUCATION

2022–Present	PhD (Quantum Machine Learning) · IIT (BHU), Varanasi 
2021-22	Doctoral Research Assistant · NIT Patna 
2018-20	M.S. Mathematical Physics · NIT Arunachal Pradesh 
2012-16	B.Tech Mechanical Engineering · (U.T.U.) 

CORE SKILLS

Python	
PyTorch	
Tensorflow	
Qiskit	
PennyLane	
LaTeX	

SELECTED PUBLICATIONS / PREPRINTS

2026	<i>HQNN-FSP: Hybrid Quantum Neural Network for Financial Stock Prediction.</i> (Quantum Mach. Intell. 8, 55 (2026). https://doi.org/10.1007/s42484-026-00390-9)
2025	<i>Graph-Based Bayesian Optimization of Variational Quantum Circuits using GNN Surrogates.</i> (arXiv: https://doi.org/10.48550/arXiv.2512.0958)

CONFERENCES

2024	Revolutionizing Drug Delivery and Screening through Quantum Machine Learning
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CERTIFICATES & SERVICE

2025	Peer Review Certificate (International Journal of Finance)
2023	Quantum Information and Quantum Technology 2023 by IHUB Quantum Technology Foundation/IISERKolkata
2023	Quantum Computing using Indigenous Quantum Simulator QSim by Center for Development of Advanced Computing
2023	Qworld QIntern2023
2021	Quantum Computing-IBM
2019	Artificial Intelligence and Machine Learning- ATAL

HIGHLIGHTS

- Quantum Research engineer internship for 3 months in QuantumEx South Korea on Quantum Machine Learning in finance (developed quantum models/algorithms for prediction of stock market) in 2024.
- Built BO+VQC pipelines with uncertainty-aware surrogate modeling.
- Comfortable with research tooling: Colab, HPC/SLURM, Git workflows.

LANGUAGES

Hindi	Native	mother tongue
English	Professional	

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